

Il grafo di Facebook

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Social Computing – Lezione 10b

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Riassunto

- 1. Intro a Social Computing
- 2. Intro a Social Media
 - Termini, Definizioni, Esempi, Classificazioni, Storia, Statistiche
- 3. Grafi e Reti
 - Definizioni di base, Rete = Grande grafo nel mondo reale
- 4, 5, 6. Misure delle reti
 - Centralità, transitività, reciprocità, degree distribution, coefficiente di clustering, distanza, ...
- 7, 8, 9. Modelli delle reti
 - Regolari, casuali, WSSW, Scale free (e varianti)
- 10a. Comunità (cenni)

Lessons learned

- Misure
 - Data una rete, abbiamo varie grandezze da associare
 - Centralità, Clustering, Distribuzione dei gradi, Componente gigante, Distanze,...
- Modelli
 - Possiamo creare reti artificiali con (alcune) delle proprietà delle reti reali
 - "Humbled by the complexity of these networks and the difficulty of independently analyzing each one of these friendships, we **try to** design models that generate, ~~on a smaller scale,~~ graphs similar to real-world networks"
[Zafarani et al.]
- Possiamo usarli per studiare e classificare le reti

Oggi

- Studio sperimentale del grafo delle amicizie di facebook
 - Nodi = utenti
 - Archi = relazioni di "amicizia"
- 3 lavori
 - 1. 2011
 - 2. 2012
 - 3. (post su blog) 2016
- "Lettura guidata"
 - Con quello che abbiamo studiato finora siamo in grado di capire alcuni lavori scientifici che analizzano in pratica il grafo (la rete) di facebook

Lavoro 1

- (Nel 2011)

arXiv:1111.4503v1 [cs.SI] 18 Nov 2011

The Anatomy of the Facebook Social Graph

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Abstract

We study the structure of the social graph of active Facebook users, the analyzed. We compute numerous features of the graph including the number degree distribution, path lengths, clustering, and mixing patterns. Our resul observations. First, we characterize the global structure of the graph, determi is nearly fully connected, with 99.91% of individuals belonging to a single and we confirm the ‘six degrees of separation’ phenomenon on a global s the average local clustering coefficient and degeneracy of graph neighborhoo Facebook graph as a whole is clearly sparse, the graph neighborhoods of user structure. Third, we characterize the assortativity patterns present in the g demographic and network properties of users. We observe clear degree assort extent to which ‘your friends have more friends than you’. Furthermore, w age on friendship preferences as well as a globally modular community stru but we do not find any strong gender homophily. We compare our results wi networks and find mostly, but not entirely, agreement on common structural

Introduction

The emergence of online social networking services over the past decade ha scientists study the structure of human relationships [1]. As individuals bring the focal point of the internet is evolving from being a network of documents t and previously invisible social structures are being captured at tremendous sc detail. In this work, we characterize the structure of the world’s largest onlin in an effort to advance the state of the art in the empirical study of social n

In its simplest form, a social network contains individuals as vertices and ec vertices [2]. This abstract view of human relationships, while certainly limit

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Abstract

We study the structure of the social graph of active Facebook users, the largest social network ever analyzed. We compute numerous features of the graph including the number of users and friendships, the degree distribution, path lengths, clustering, and mixing patterns. Our results center around three main observations. First, we characterize the global structure of the graph, determining that the social network is **nearly fully connected**, with 99.91% of individuals belonging to a **single large connected component**, and we confirm the **'six degrees of separation'** phenomenon on a global scale. Second, by studying the **average local clustering coefficient** and degeneracy of graph neighborhoods, we show that while the Facebook graph as a whole is clearly sparse, the graph neighborhoods of users contain surprisingly dense structure. Third, we characterize the assortativity patterns present in the graph by studying the basic demographic and network properties of users. We observe clear degree assortativity and characterize the extent to which **'your friends have more friends than you'**. Furthermore, we observe a strong effect of age on friendship preferences as well as a globally modular **community structure driven by nationality**, but we do not find any strong gender homophily. We compare our results with those from smaller social networks and find mostly, but not entirely, agreement on common structural network characteristics.

Il Grafo Sociale di Facebook

- Nodi:
 - Utenti
 - (utenti attivi: login negli ultimi 28 gg)
 - 721 milioni, > 10% popolazione mondiale
- Archi
 - "Amicizia" su FB
 - ~69 miliardi → ca 190 amici in media
- Sottografo U.S.
 - 149 milioni di utenti
 - ~16 miliardi di archi → ca 210 amici in media

Facebook degree distribution

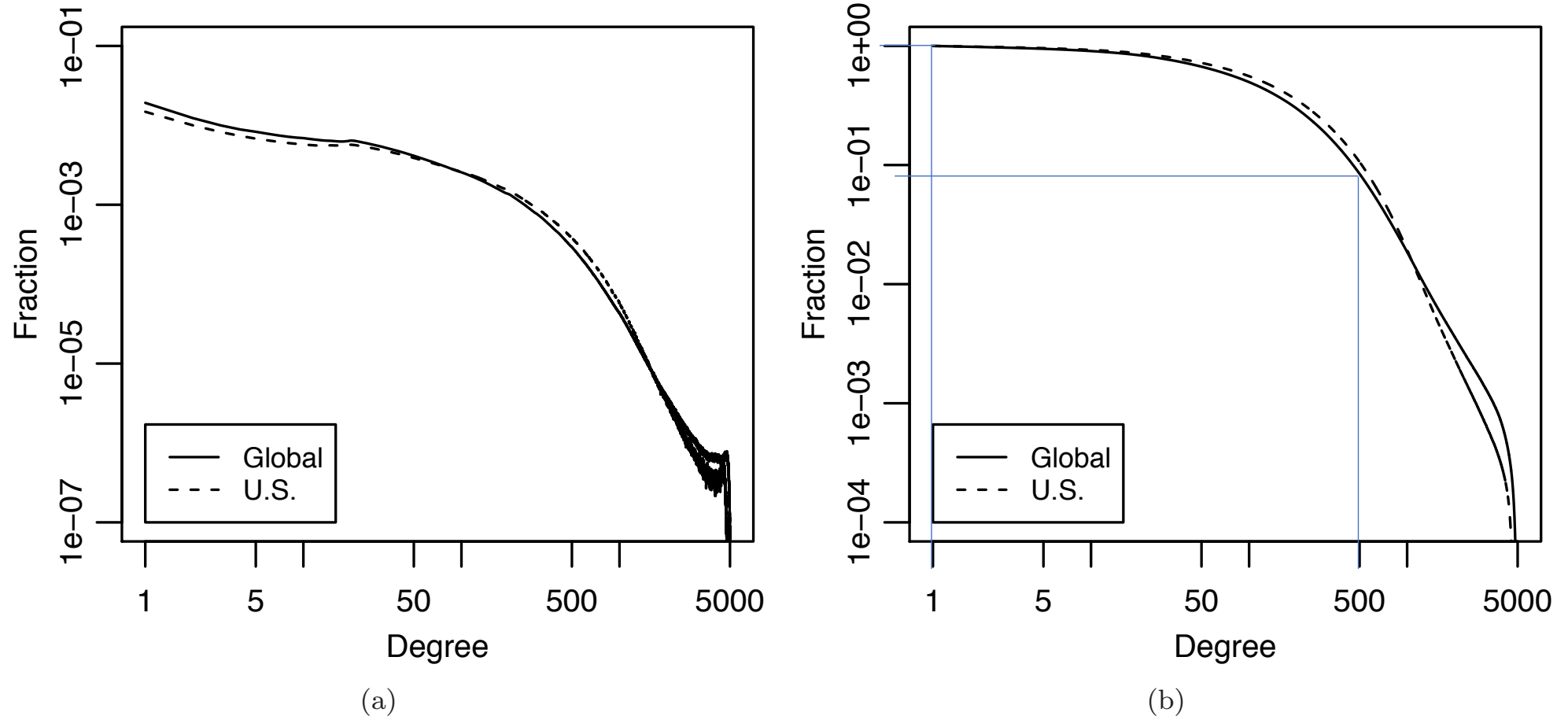


Figure 1. Degree distribution p_k . (a) The fraction of users with degree k for both the global and U.S. population of Facebook users. (b) The complementary cumulative distribution function (CCDF). The CCDF at degree k measures the fraction of users who have degree k or greater and in terms of the degree distribution is $\sum_{k' \geq k} p_{k'}$. For the U.S., the degree measures the number of friends also from the United States.

Commenti

- $\sim$ globale e U.S.
 - U.S. un po' più diffuso all'epoca
- Monotona decrescente
 - (picco a 20, politiche di Facebook)
- Cutoff a 5000
 - Limite imposto da Facebook
- Assomiglia un po' a una power-law, ma non è proprio power-law...
 - (Sembrano "2 power law")
- Con la CCDF si vede bene come il rumore nella coda scompare

Facebook components size

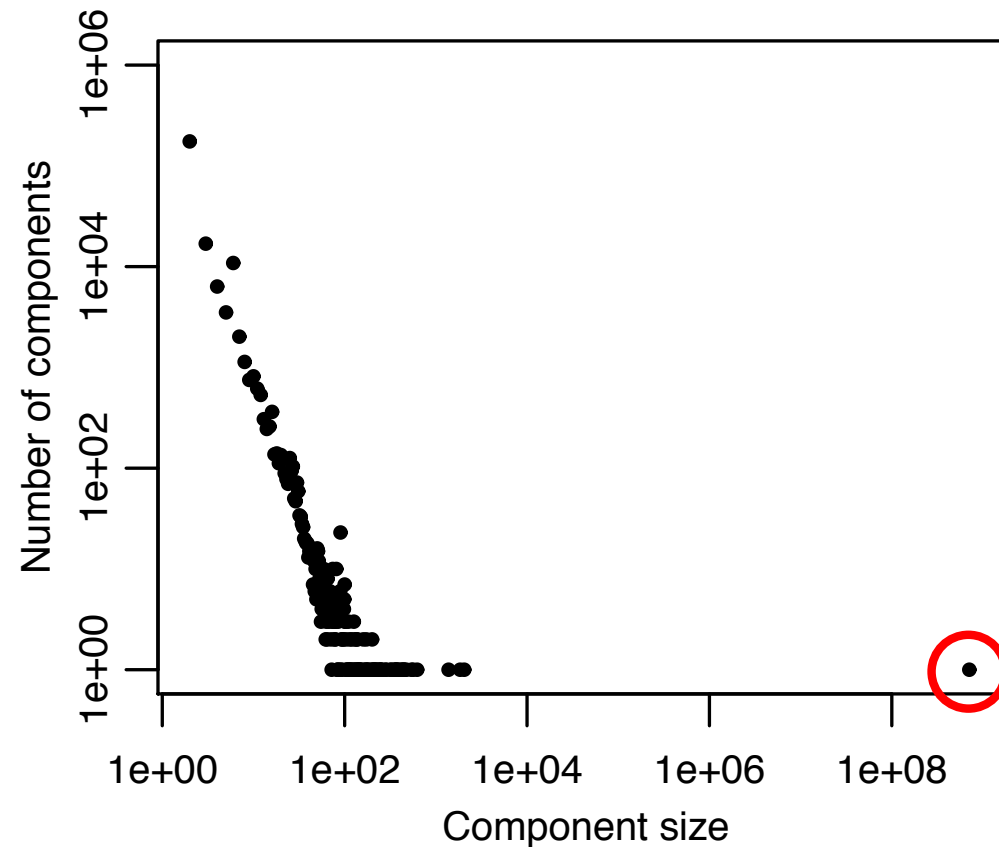
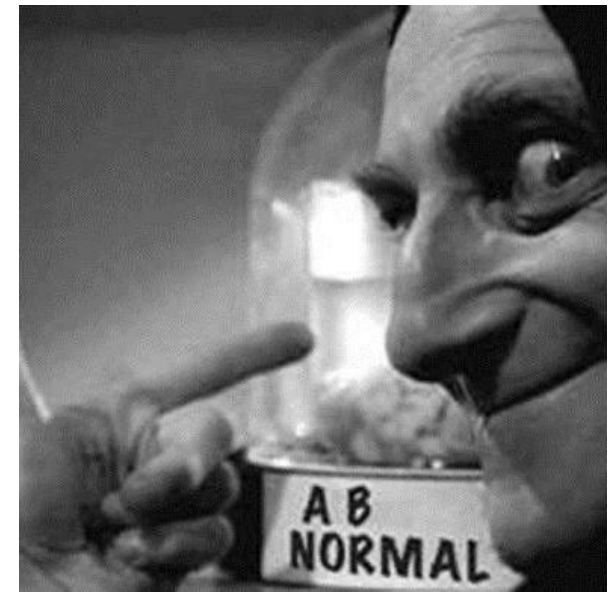


Figure 3. Component size distribution. The fraction of components with a given component size on a log-log scale. Most vertices (99.91%) are in the largest component.

Commenti

- Questa sì è power-law
 - E, se la distribuzione della grandezza delle componenti è una power-law => c'è una componente di dimensione "abnorme"
- C'è una "Giant component"
 - (la seconda componente ha solo 2000 nodi...)
- Quindi
 - Quasi tutti i nodi sono connessi
 - Ha senso parlare di diametro e distanza media
- N.B. Qui si vede anche bene perché la media su una power law ha poco senso
 - La dimensione media sarà intorno al 100, ma le cose interessanti sono altrove!



Facebook average distance

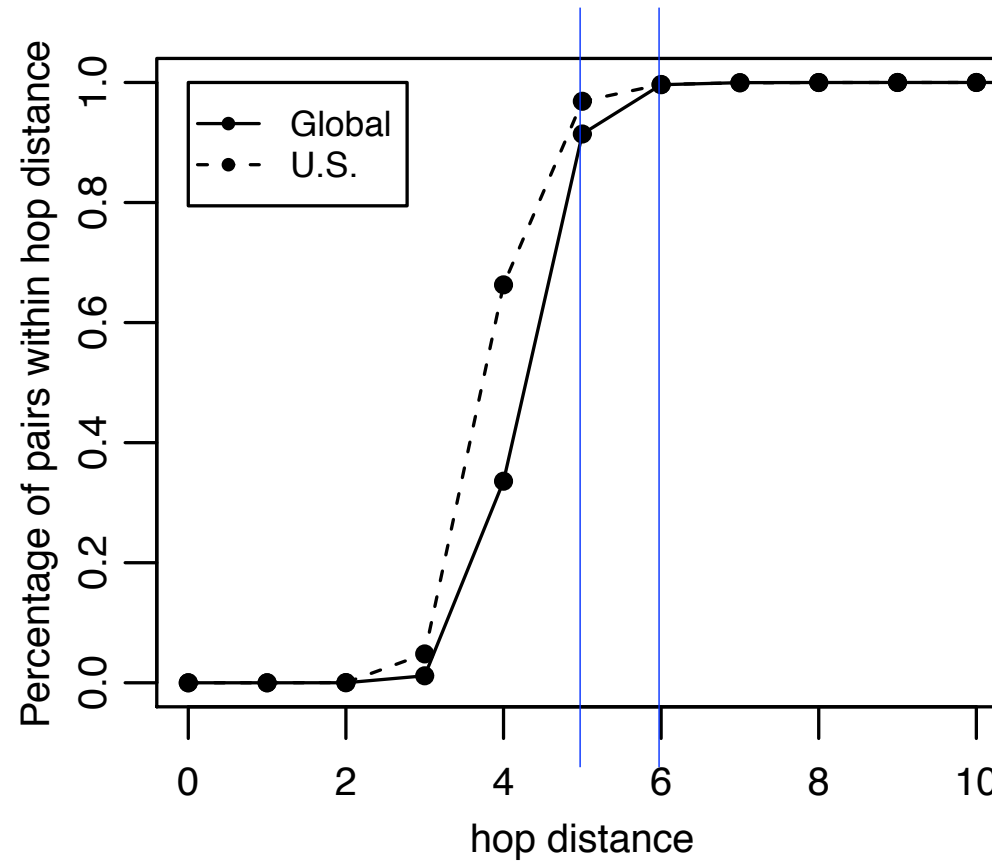


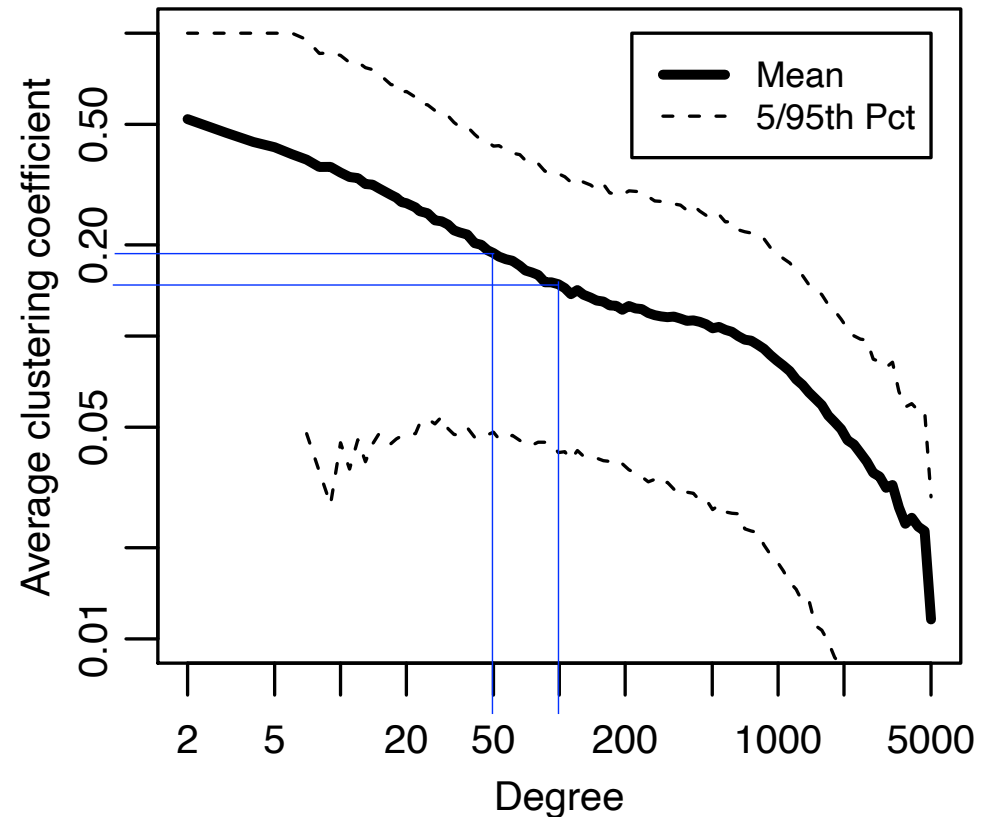
Figure 2. Diameter. The neighborhood function $N(h)$ showing the percentage of user pairs that are within h hops of each other. The average distance between users on Facebook in May 2011 was 4.7, while the average distance within the U.S. at the same time was 4.3.

Mondo piccolo

- 92%: meno di 5 gradi di separazione
- 99.6%: meno di 6
- U.S.: ancora più piccolo
- Diametro 🙄🙈

Coefficiente di clustering (C)

- In funzione del grado
- È alto. Ad es.
 - $\sim .2$ per grado 50
 - $\sim .15$ per grado 100
- Monotono decrescente
 - (normale: se grado alto per avere lo stesso C serve che i suoi vicini abbiano grado alto)
 - Decrescita rapida a 5000 → profili con altri scopi, gruppo di amici non "coerente"



Paradosso degli amici

- "I tuoi amici hanno più amici di te"
- Vero per chi ha meno di 700 amici

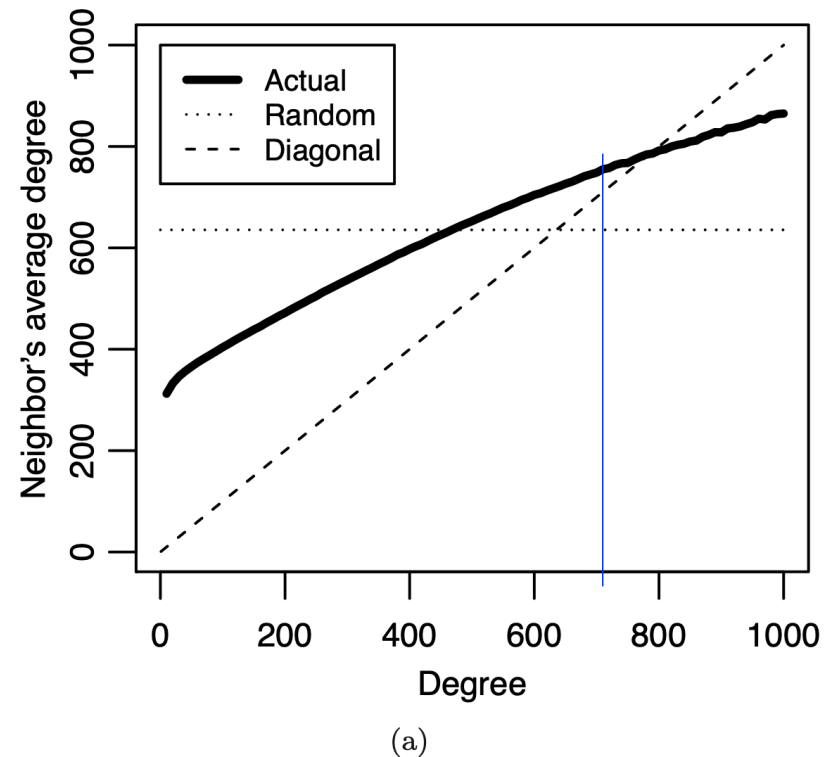


Figure 6. Degree correlations. (a) The average r

Paese di appartenenza (Comunità $\leftrightarrow$ Nazionalità)

- (Sulla base dell'IP)
- 84.2% degli archi sono all'interno di un paese
 - Intuitivo, si hanno più "amici" del proprio paese
 - (diagonale colori più chiari)
- Ma ci sono anche gruppi di paesi
 - (quadrati con colori più chiari)
 - Sulla base di lingua, cultura, posizione...
 - Es.:
 - AU, NZ, GB
 - Paesi latini di lingua spagnola/portoghese
 - DK, NO, SE
 - ...

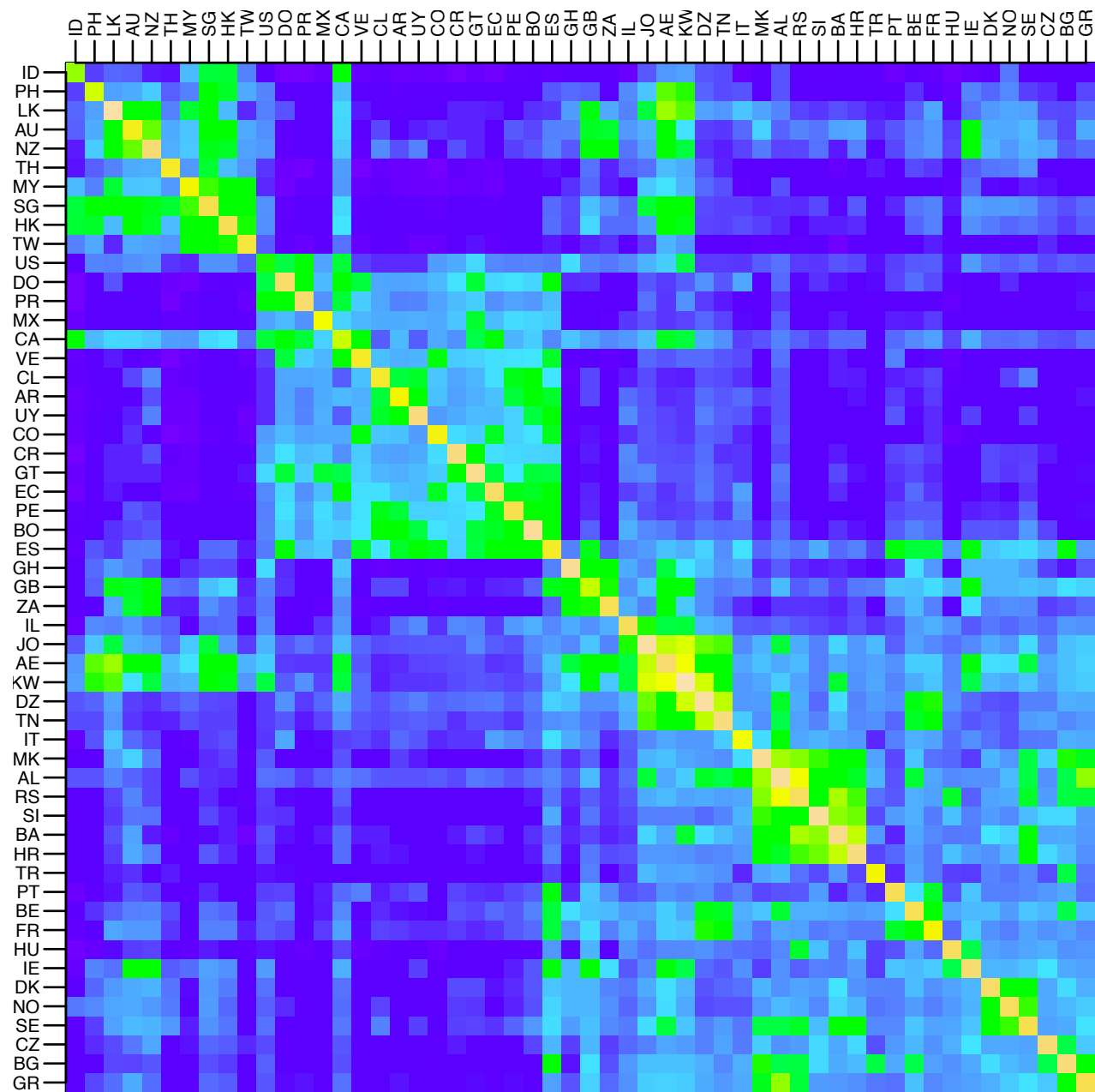


Figure 9. Normalized country adjacency matrix. Matrix of edges between countries with > 1 million users and $> 50\%$ Facebook penetration shown on a log scale. To normalize, we divided each element of the adjacency matrix by the product of the row country degree and column country degree.

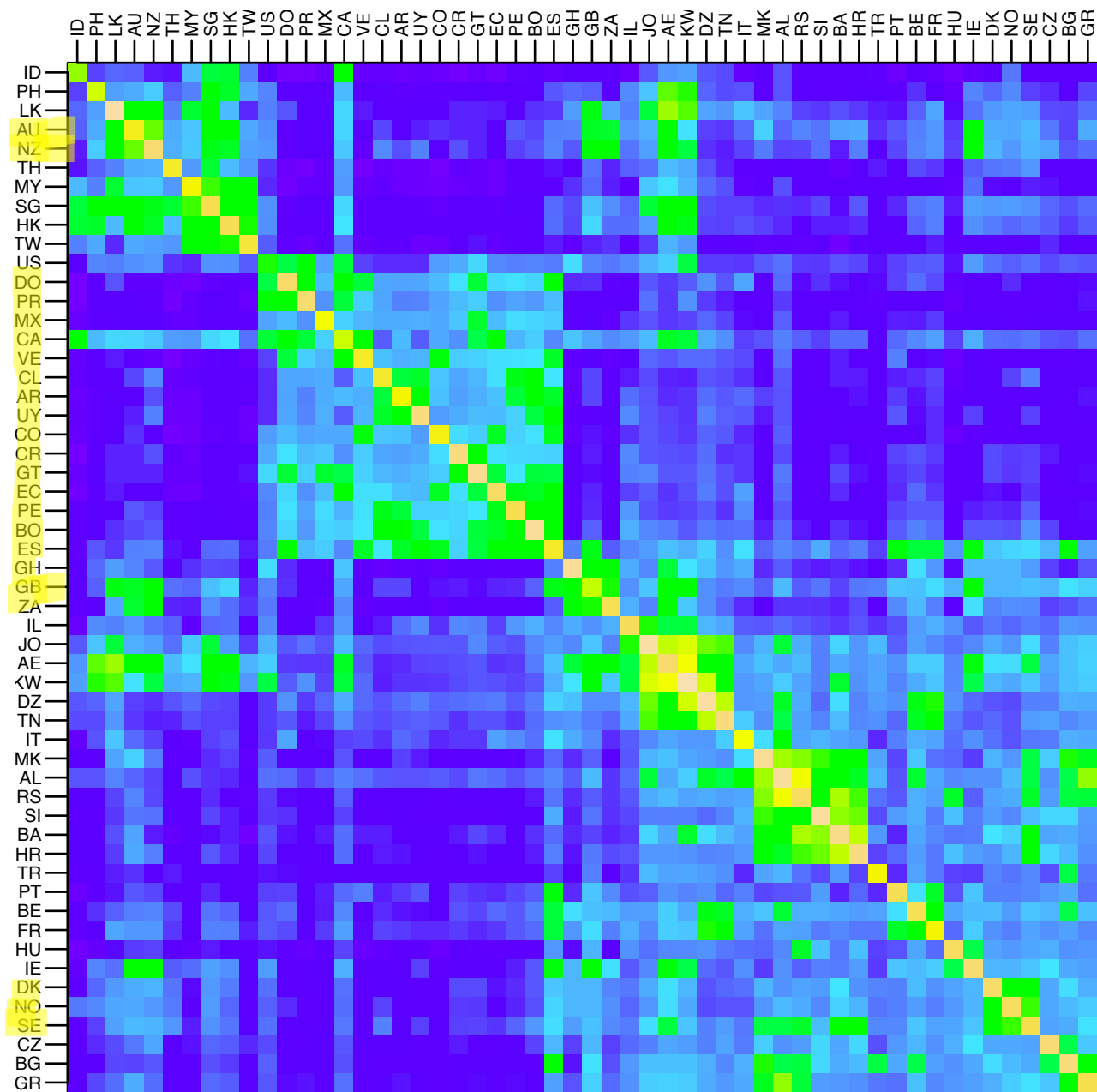


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Lavoro 2

- Nel 2012 (1 anno dopo)
- Sempre sullo stesso grafo
- Alcune differenze:
 - Soprattutto distanza
 - Evoluzione temporale
 - Sottografi
 - us
 - it
 - se
 - itse

Four Degrees of Separation

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ABSTRACT

Frigyés Karinthy, in his 1929 short story “Láncszemek” (in English, “Chains”) suggested that any two persons are distanced by at most six friendship links.¹ Stanley Milgram in his famous experiments challenged people to route postcards to a fixed recipient by passing them only through direct acquaintances. Milgram found that the average number of intermediaries on the path of the postcards lay between 4.4 and 5.7, depending on the sample of people chosen. We report the results of the first world-scale social-network graph-distance computations, using the entire Facebook network of active users (≈ 721 million users, ≈ 69 billion friendship links). The average distance we observe is 4.74, corresponding to 3.74 intermediaries or “degrees of separation”, prompting the title of this paper. More generally, we study the distance distribution of Facebook and of some interesting geographic subgraphs, looking also at their evolution over time. The networks we are able to explore are almost two orders of magnitude larger than those analysed in the previous literature. We report detailed statistical metadata showing that our measurements (which rely on probabilistic algorithms) are very accurate.

INTRODUCTION

*Paolo Boldi, Marco Rosa and Sebastiano Vigna have been partially supported by a Yahoo! faculty grant and by MIUR PRIN “Query log e web crawling”. This research is done in the frame of the EC FET NADINE project.

¹The exact wording of the story is slightly ambiguous: “He bet us that, using no more than five individuals, one of whom is a personal acquaintance, he could contact the selected individual [...]”. It is not completely clear whether the selected individual is part of the five, so this could actually allude to distance five or six in the language of graph theory, but the “six degrees of separation” phrase stuck after John Guare’s 1990 eponymous play. Following Milgram’s definition and Guare’s interpretation (see further on), we will assume that “degrees of separation” is the same as “distance minus one”, where “distance” is the usual path length (the number of arcs in the path).

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At the 20th World-Wide Web Conference, in Hyderabad, India, one of the authors (Sebastiano) presented a new tool for studying the distance distribution of very large graphs: HyperANF [3]. Building on previous graph compression work [4] and on the idea of diffusive computation pioneered in [19], the new tool made it possible to accurately study the distance distribution of graphs orders of magnitude larger than what was previously possible.

One of the goals in studying the distance distribution is the identification of interesting statistical parameters that can be used to tell proper social networks from other complex networks, such as web graphs. More generally, the distance distribution is one interesting *global* feature that makes it possible to reject probabilistic models even when they match local features such as the in-degree distribution. In particular, earlier work [3] had shown that the *spid*², which measures the *dispersion* of the distance distribution, appeared to be smaller than 1 (underdispersion) for social networks, but larger than one (overdispersion) for web graphs. Hence, during the talk, one of the main open questions was “What is the spid of Facebook?”.

Lars Backstrom happened to listen to the talk, and suggested a collaboration studying the Facebook graph. This was of course an extremely intriguing possibility: beside testing the “spid hypothesis”, computing the distance distribution of the Facebook graph would have been the largest Milgram-like [18] experiment ever performed, orders of magnitudes larger than previous attempts (during our experiments Facebook has ≈ 721 million active users and ≈ 69 billion friendship links).

This paper reports our findings in studying the distance distribution of the largest electronic social network ever created. The average distance of the current Facebook graph is 4.74. Moreover, the spid of the graph is just 0.09, corroborating the conjecture [3] that proper social networks have a spid well below one. Contrary to what has been commonly observed analysing graphs orders of magnitude smaller, we also observe both a stabilisation of the average distance over time and that the density of the graph over time does not neatly fit previous models. Towards a deeper understanding of the structure of the Facebook graph, we apply recent compression techniques that exploit the underlying cluster

²The spid (shortest-paths index of dispersion) is the variance-to-mean ratio of the distance distribution.

nodi, # archi

- Chiaramente in crescita con il passare del tempo
- (current = 2012)

	it	se	itse	us	fb
2007	159.8 K (105.0 K)	11.2 K (21.8 K)	172.1 K (128.8 K)	8.8 M (529.3 M)	13.0 M (644.6 M)
2008	335.8 K (987.9 K)	1.0 M (23.2 M)	1.4 M (24.3 M)	20.1 M (1.1 G)	56.0 M (2.1 G)
2009	4.6 M (116.0 M)	1.6 M (55.5 M)	6.2 M (172.1 M)	41.5 M (2.3 G)	139.1 M (6.2 G)
2010	11.8 M (726.9 M)	3.0 M (149.9 M)	14.8 M (878.4 M)	92.4 M (6.0 G)	332.3 M (18.8 G)
2011	17.1 M (1.7 G)	4.0 M (278.2 M)	21.1 M (2.0 G)	131.4 M (12.4 G)	562.4 M (47.5 G)
current	19.8 M (2.2 G)	4.3 M (335.7 M)	24.1 M (2.6 G)	149.1 M (15.9 G)	721.1 M (68.7 G)

Table 5. Number of nodes and friendship links of the datasets. Note that each friendship link, being undirected, is represented by a pair of symmetric arcs.

Studio precedente

Grado medio: variazione temporale

- Chiaramente in crescita con il passare del tempo
- (current = 2012)

	it	se	itse	us	fb
2007	1.31	3.90	1.50	119.61	99.50
2008	5.88	46.09	36.00	106.05	76.15
2009	50.82	69.60	55.91	111.78	88.68
2010	122.92	100.85	118.54	128.95	113.00
2011	198.20	140.55	187.48	188.30	169.03
current	226.03	154.54	213.30	213.76	190.44

Table 2. Average degree of the datasets.

nodi vs. # archi

- Crescono entrambi

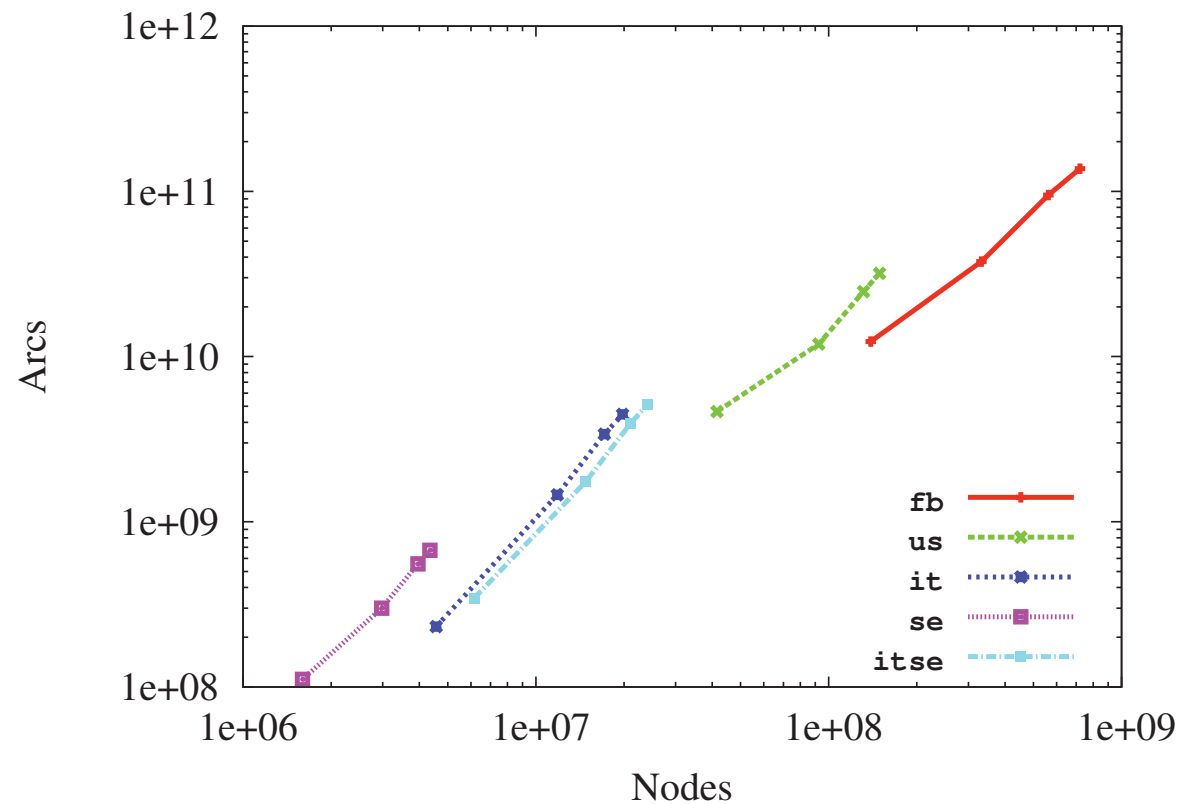


Figure 6. A plot correlating number of nodes to the average degree (for the graphs from 2009 on).

Ma la densità cala

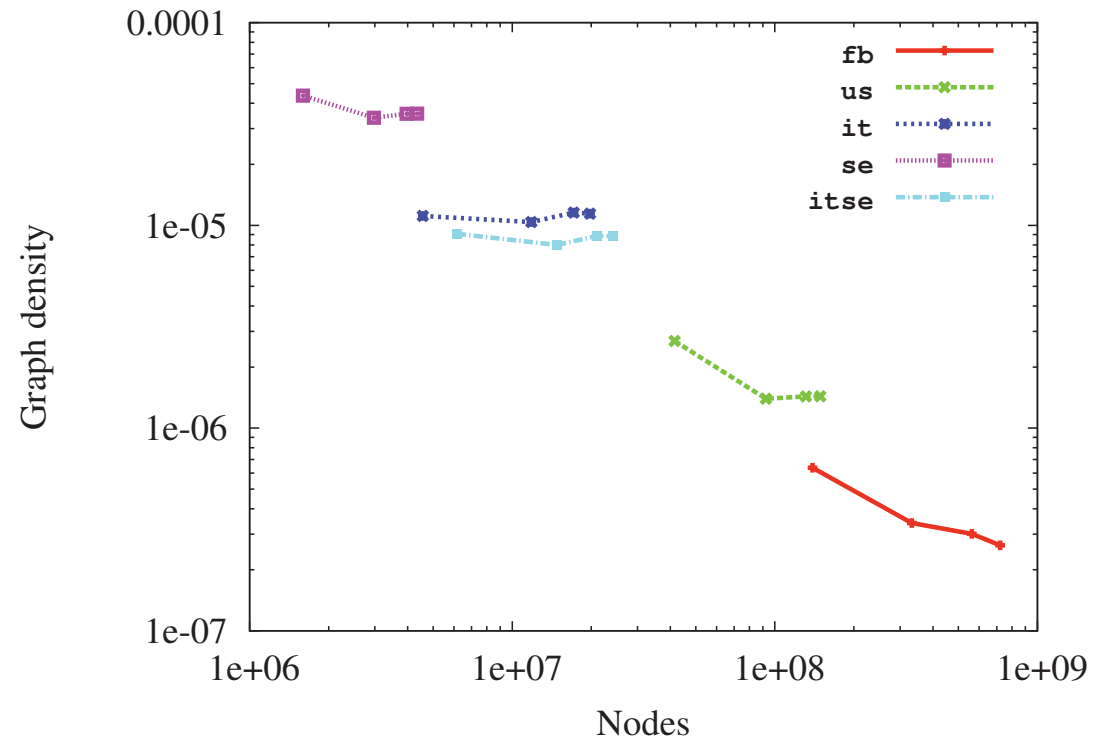


Figure 5. A plot correlating number of nodes to graph density (for the graph from 2009 on).

- Densità = $\# \text{ archi} / \# \text{ archi possibili}$
- Più alta per porzioni: probabilmente perché ci sono altre porzioni "immature" (con nodi "mancanti")

Distanze

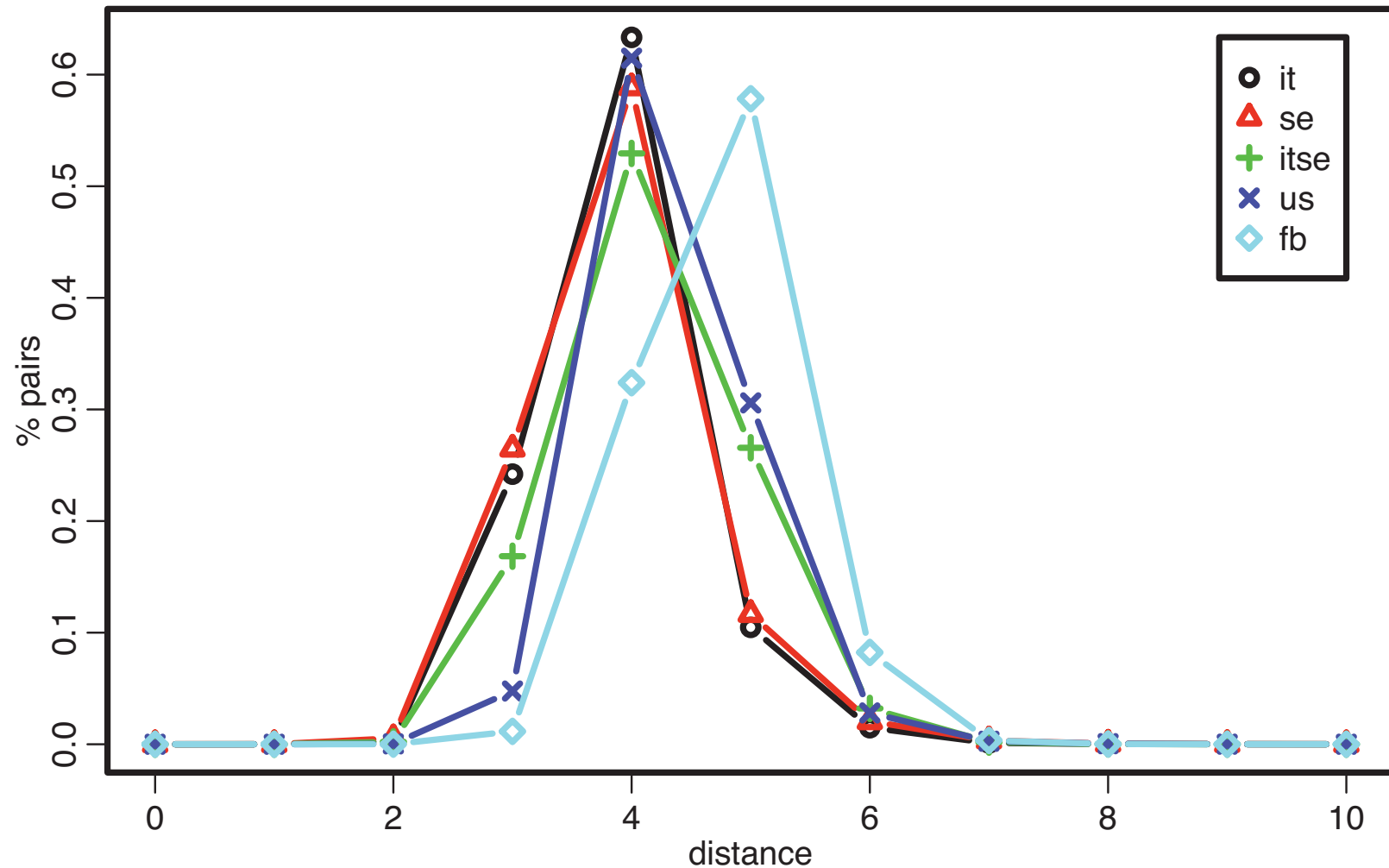
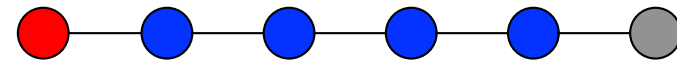


Figure 2. The probability mass functions of the distance distributions of the current graphs (truncated at distance 10).

"Tutto fb" vs. Regionale

- **Medie**
 - Tutto fb: ~5
 - Regionali (it, se, itse, us): ~4
- **N.B. Reachability:**

«people are in fact only four worlds apart, and not six: when considering another person in the world, a friend of your friend knows a friend of their friend, on average»
[ultima frase dell'articolo] (e titolo)



	it	se	itse	us	fb
2007	0.04	10.23	0.19	100.00	68.02
2008	25.54	93.90	80.21	99.26	89.04

Table 3. Percentage of reachable pairs 2007–2008.

Convergenza rapida dopo 2008

- Varianza cala

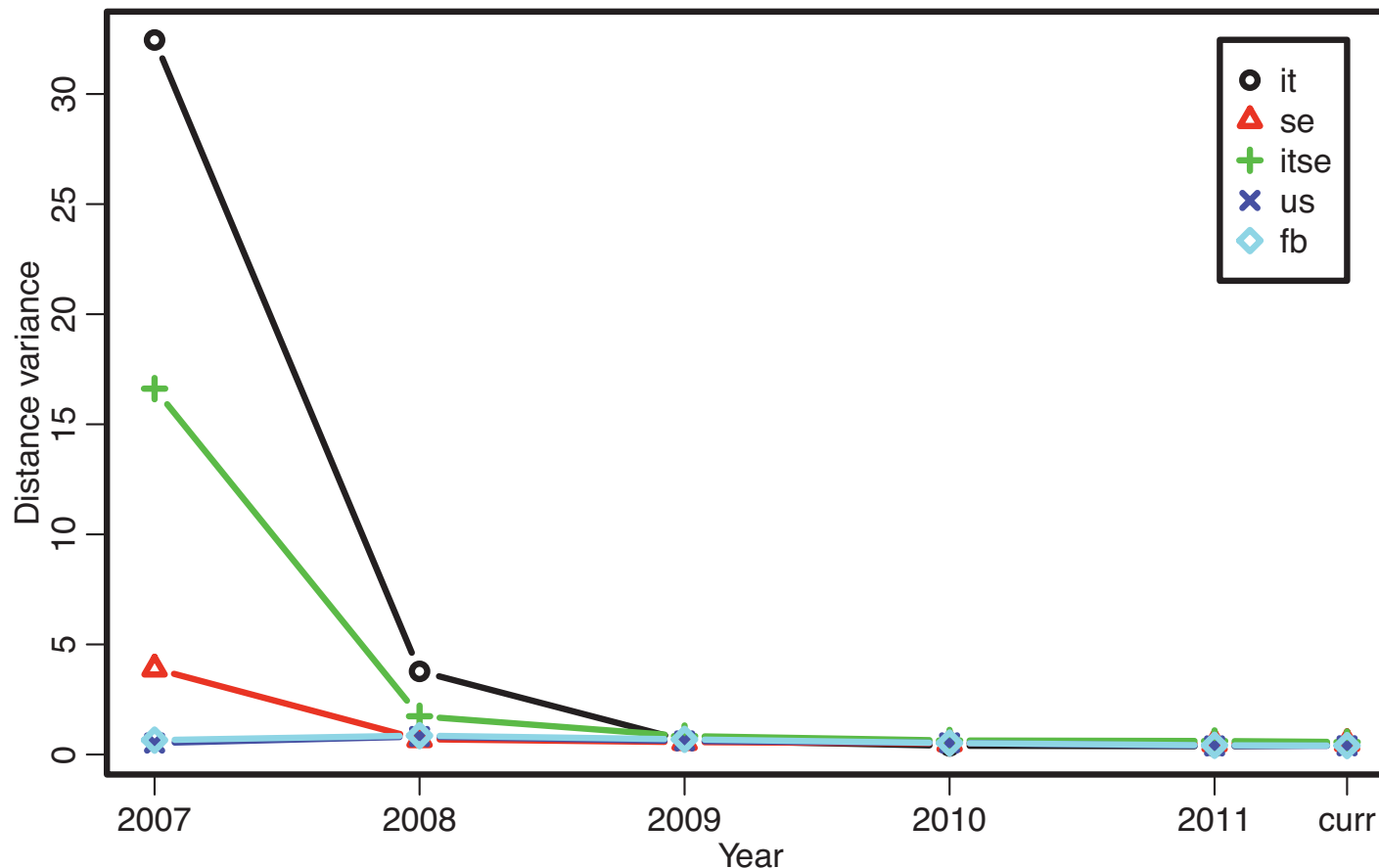


Figure 4. The graph of variances of the distance distributions. See also Table 7.

Distanza media stabile dopo 2008

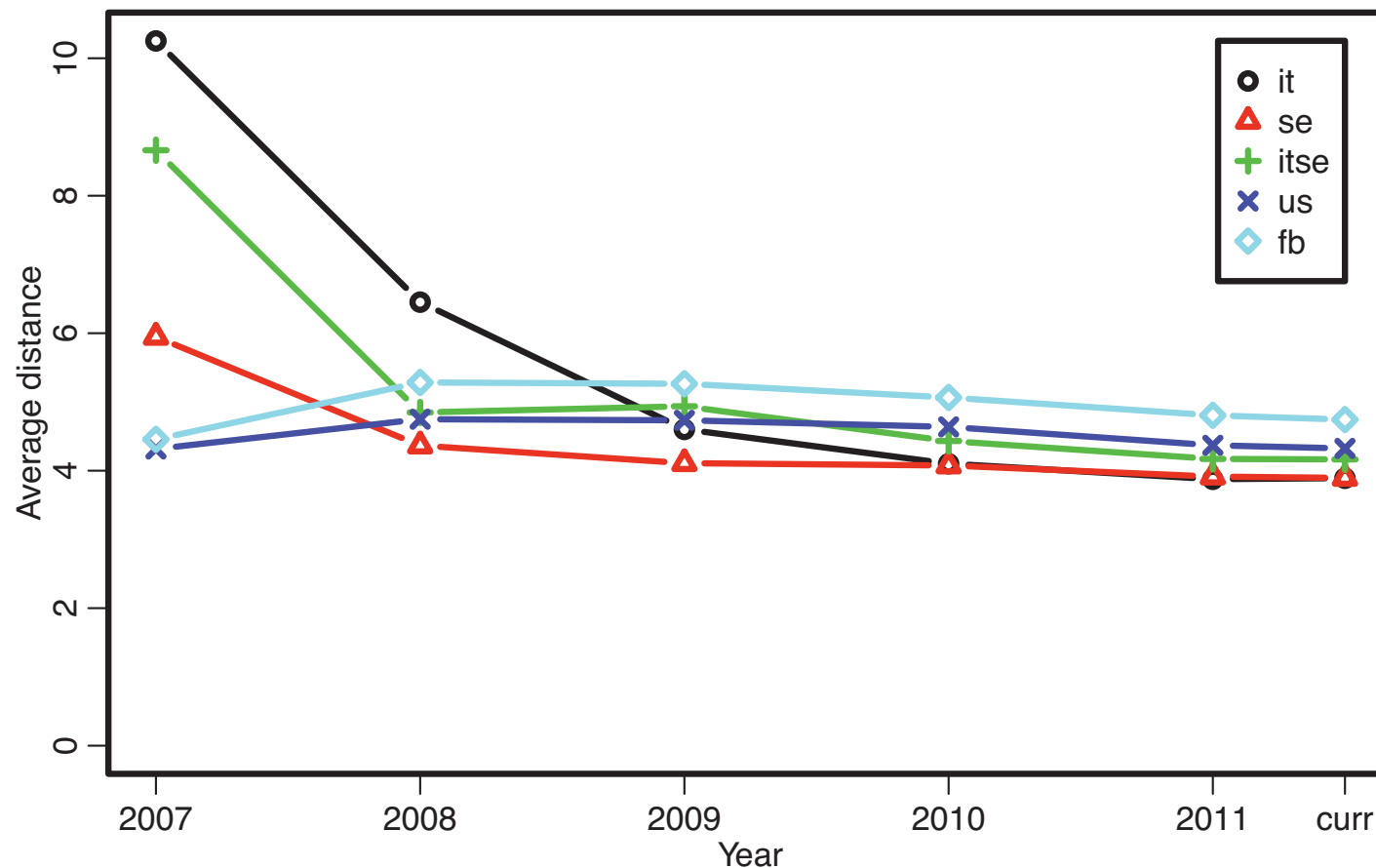
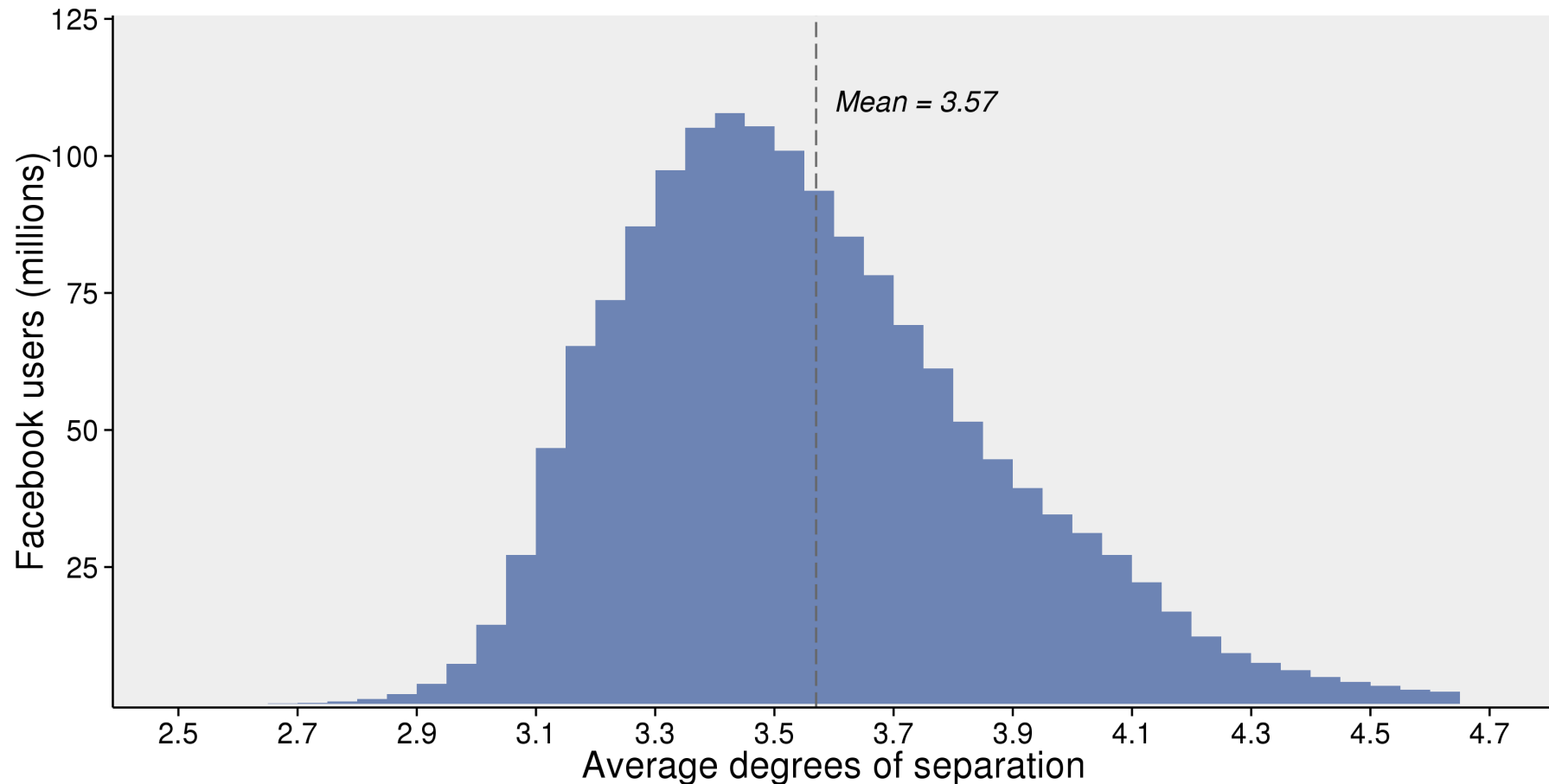


Figure 3. The average distance graph. See also Table 6.

"Lavoro" 3 [2016]: distanza = 3.5



Blog post su facebook research blog

<https://research.fb.com/blog/2016/02/three-and-a-half-degrees-of-separation/>

Riassunto

- Studi sperimentali del grafo di Facebook
- Non semplice, problemi computazionali...
 - (vero in generale)
- Caratteristiche:
 - Non proprio power-law
 - Connesso
 - C alto
 - Piccolo mondo (distanze basse)
- Andamento temporale
 - Connettività cresce
 - Distanze e densità calano
- Ora avete le conoscenze per **capire** gli studi
- Prossime lezioni (di "lab"): conoscenze per **"fare"**

Biblio

- I 2 – 3 lavori:

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